

Most of football analysis continues to rely on outcome-proximate statistics such as expected goals to assess the value of a player and the silent methodology is to ignore any player whose value doesn't appear in that column. Jiang, Cai, and Kyrillidis (2025) approach this head-on, they define individual credit for change in expected threat in a graph of player interactions, and experiment with variants of both Graph Attention Network and Transformer models to identify "hidden pivotal players" the deep-lying midfielders and ball-playing defenders who rarely appear in a highlight. Even though it works but it is essentially a black box as if a coach observes a player's score, he cannot tell specifically what motivated it.

Bischofberger et al. (2026) tackle a more limited but even more specific question on defensive off-ball value. They create Defensive Pressure Areas and role-conditioned baselines from full player tracking data and then compare and contrast the changes in expected threats to individual players with market values and external ratings. While it's a decent contribution, it's only defense and it lacks tracking metrics that very few clubs possess. The attacking counterpart is proposed by Fernández, Bornn, and Cervone (2019, 2021), which uses their Expected Possession Value model to assign a value to a possession that's already occurred, but also to the current position of players and their direction of play. It's similar to the defensive work above as it also relies upon tracking data, which is often not public, is not cheap, and simply doesn't exist at a lower level in the game. VAEP (Decroos, Bransen, Van Haaren, & Davis, 2019) underlies all of this and treats any on-ball action whether it's a pass, a tackle, a clearance as a move that moves the probability of scoring/losing a goal. VAEP is based on the events themselves (encoded in the open source socceraction library), but it can only award the player who is actually playing with the ball. The scoring goal that gets the defender out of position, the positioning that enables a pass, all that is absent.

We suggest the Aura Score, an interpretable model for the catalytic contribution, or the value added by the player's actions and positioning in the following moments, both in attack and defence, without the need for constant player tracking data. In contrast to VAEP's limited binary labels, we predict cumulative VAEP value that is generated over n future actions in a chain during a possession, where n is a hyperparameter and is not fixed arbitrarily, but chosen optimally via cross validation. We use the World Cup 2022 and Euro 2020 positional snapshots (more detailed than just the events, but not full tracking streams) from StatsBomb 360 to train a Random Forest and apply SHAP, as in Tsai et al. (2026), and use a fully held-out tournament (Euro 2024) to test, instead of assuming that our explanations generalize.

Reference:

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- Bischofberger et al. (2026). Blame is Easier Than Praise: Measuring Off-Ball Defensive Performance in Football. <https://arxiv.org/abs/2606.19931>
- Fernández, J., Bornn, L., & Cervone, D. (2019/2021). Decomposing the Immeasurable Sport: A Deep Learning Expected Possession Value Framework for Soccer. MIT Sloan 2019 / Machine Learning, 110(6). https://www.lukebornn.com/papers/fernandez_ssac_2019.pdf

Tsai, Y.-F. et al. (2026). Interpretable Machine Learning for Football Performance Analysis: Evidence of Limited Transferability from Elite Leagues to University Competition.
<https://arxiv.org/abs/2605.10796>